

Assignment_4

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Necessary packages

```
library(factoextra)
```

```
## Loading required package: ggplot2
```

```
## Welcome to factoextra!
```

```
## Want to learn more? See two factoextra-related books at https://www.datanovia.com/en/product/practical-guide-to-principal-component-methods-in-r/
```

```
library(cluster)
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.2.0      v readr      2.1.6
```

```
## v forcats    1.0.1      v stringr    1.6.0
```

```
## v lubridate  1.9.5      v tibble     3.3.1
```

```
## v purrr      1.2.1      v tidyr      1.3.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

Load Data

```
Pharmaceuticals_data<-read.csv("~/Desktop/R documents/Assignment files/FML_Assignment4/Pharmaceuticals.csv")
```

```
head(Pharmaceuticals_data)
```

##	Symbol	Name	Market_Cap	Beta	PE_Ratio	ROE	ROA	Asset_Turnover
## 1	ABT	Abbott Laboratories	68.44	0.32	24.7	26.4	11.8	0.7
## 2	AGN	Allergan, Inc.	7.58	0.41	82.5	12.9	5.5	0.9
## 3	AHM	Amersham plc	6.30	0.46	20.7	14.9	7.8	0.9
## 4	AZN	AstraZeneca PLC	67.63	0.52	21.5	27.4	15.4	0.9
## 5	AVE	Aventis	47.16	0.32	20.1	21.8	7.5	0.6
## 6	BAY	Bayer AG	16.90	1.11	27.9	3.9	1.4	0.6
##	Leverage	Rev_Growth	Net_Profit_Margin	Median_Recommendation	Location	Exchange		
## 1	0.42	7.54	16.1	Moderate Buy	US	NYSE		
## 2	0.60	9.16	5.5	Moderate Buy	CANADA	NYSE		
## 3	0.27	7.05	11.2	Strong Buy	UK	NYSE		
## 4	0.00	15.00	18.0	Moderate Sell	UK	NYSE		
## 5	0.34	26.81	12.9	Moderate Buy	FRANCE	NYSE		
## 6	0.00	-3.17	2.6	Hold	GERMANY	NYSE		

```
summary(Pharmaceuticals_data)
```

```
##      Symbol      Name      Market_Cap      Beta
## Length:21      Length:21      Min.   : 0.41      Min.   :0.1800
## Class :character Class :character 1st Qu.: 6.30      1st Qu.:0.3500
## Mode  :character Mode  :character Median : 48.19      Median :0.4600
##                                     Mean  : 57.65      Mean  :0.5257
##                                     3rd Qu.: 73.84      3rd Qu.:0.6500
##                                     Max.   :199.47      Max.   :1.1100
##      PE_Ratio      ROE      ROA      Asset_Turnover      Leverage
## Min.   : 3.60      Min.   : 3.9      Min.   : 1.40      Min.   :0.3      Min.   :0.0000
## 1st Qu.:18.90      1st Qu.:14.9      1st Qu.: 5.70      1st Qu.:0.6      1st Qu.:0.1600
## Median :21.50      Median :22.6      Median :11.20      Median :0.6      Median :0.3400
## Mean   :25.46      Mean   :25.8      Mean   :10.51      Mean   :0.7      Mean   :0.5857
## 3rd Qu.:27.90      3rd Qu.:31.0      3rd Qu.:15.00      3rd Qu.:0.9      3rd Qu.:0.6000
## Max.   :82.50      Max.   :62.9      Max.   :20.30      Max.   :1.1      Max.   :3.5100
##      Rev_Growth      Net_Profit_Margin      Median_Recommendation      Location
## Min.   : -3.17      Min.   : 2.6      Length:21      Length:21
## 1st Qu.: 6.38      1st Qu.:11.2      Class :character      Class :character
## Median : 9.37      Median :16.1      Mode  :character      Mode  :character
## Mean   :13.37      Mean   :15.7
## 3rd Qu.:21.87      3rd Qu.:21.1
## Max.   :34.21      Max.   :25.5
##      Exchange
## Length:21
## Class :character
## Mode  :character
##
##
##
```

Selecting Only Numerical Variables (1–9)

```
pharma_numericaldata <- Pharmaceuticals_data[, 3:11]
```

```
colnames(pharma_numericaldata)
```

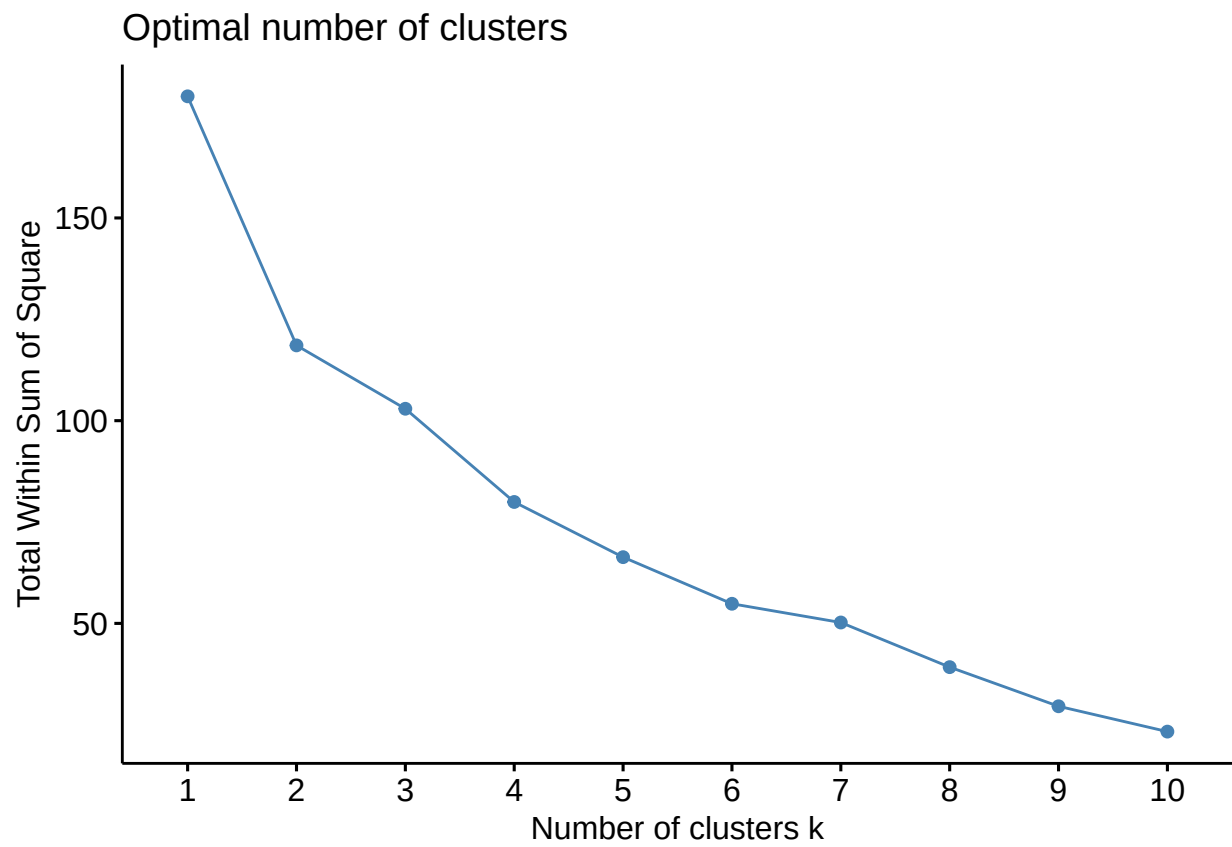
```
## [1] "Market_Cap"      "Beta"      "PE_Ratio"
## [4] "ROE"             "ROA"       "Asset_Turnover"
## [7] "Leverage"        "Rev_Growth" "Net_Profit_Margin"
```

Normalizing the data

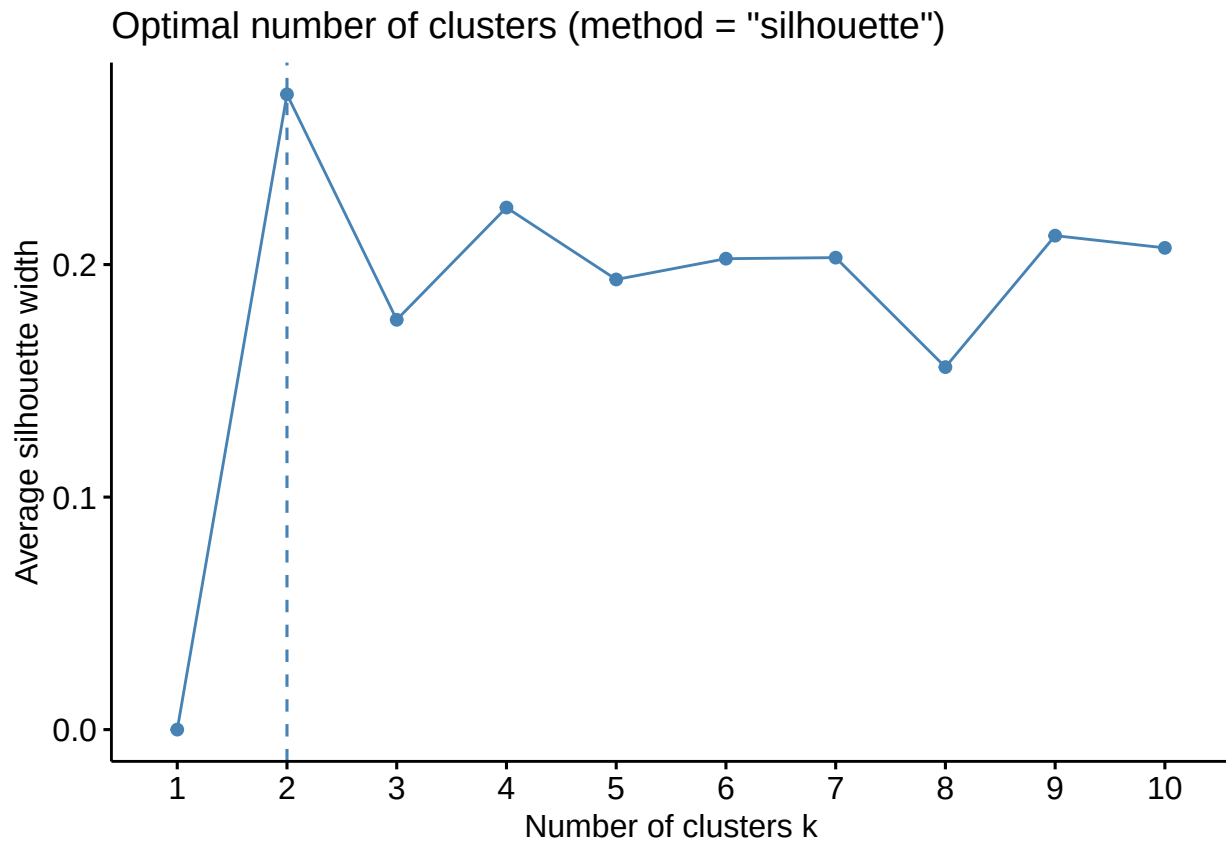
```
Pharmanumericals_normal <- scale(pharma_numericaldata)
```

Choosing number of clusters using Elbow method

```
set.seed(123)
library(factoextra)
fviz_nbclust(Pharmanumericals_normal, kmeans, method = "wss")
```



```
fviz_nbclust(Pharmanumericals_normal, kmeans, method = "silhouette")
```



The optimal k=2

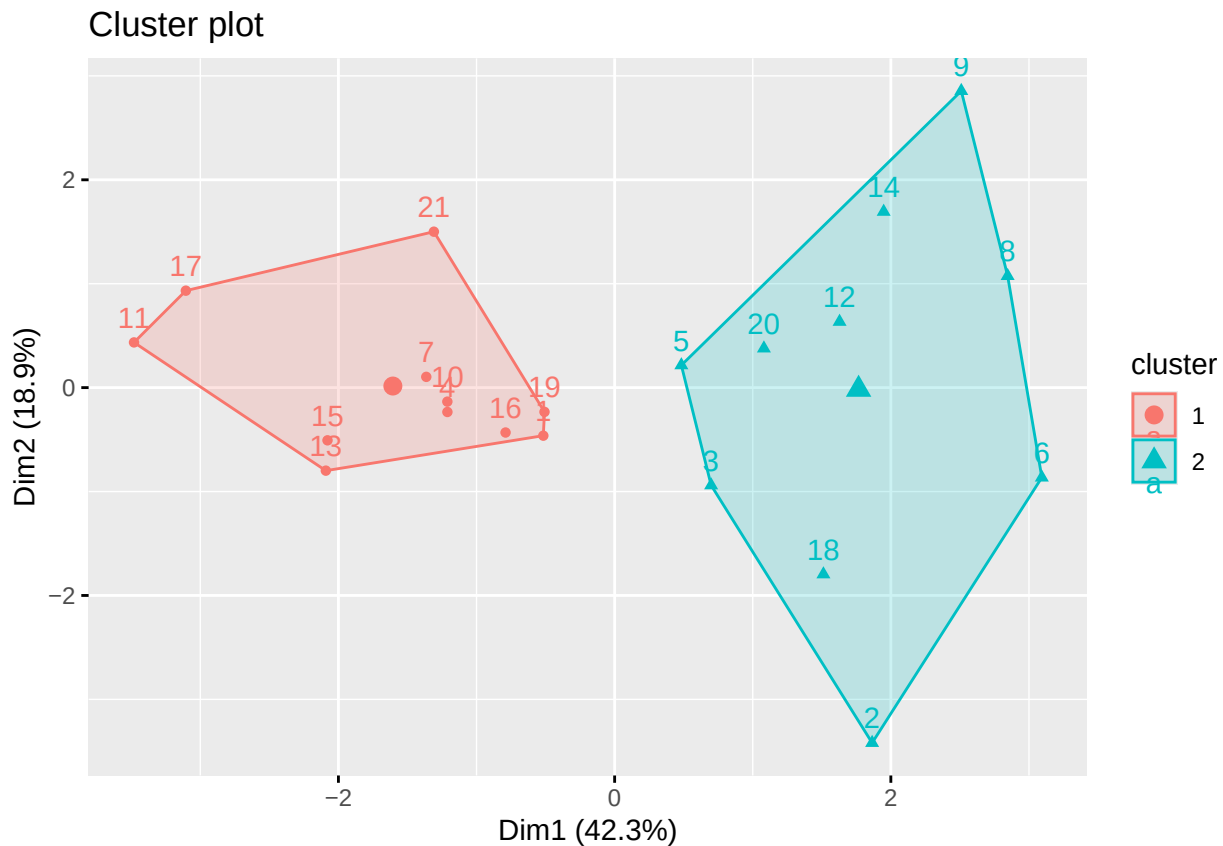
K-mean clustering

```
kmeans_result <- kmeans(Pharmanumericals_normal, centers = 2, nstart = 25)
kmeans_result
```

```
## K-means clustering with 2 clusters of sizes 11, 10
##
## Cluster means:
##   Market_Cap      Beta  PE_Ratio      ROE      ROA Asset_Turnover
## 1  0.6733825 -0.3586419 -0.2763512  0.6565978  0.8344159    0.4612656
## 2 -0.7407208  0.3945061  0.3039863 -0.7222576 -0.9178575   -0.5073922
##   Leverage Rev_Growth Net_Profit_Margin
## 1 -0.3331068 -0.2902163      0.6823310
## 2  0.3664175  0.3192379     -0.7505641
##
## Clustering vector:
## [1] 1 2 2 1 2 2 1 2 2 1 1 2 1 2 1 1 1 2 1 2 1
##
## Within cluster sum of squares by cluster:
## [1] 43.30886 75.26049
## (between_SS / total_SS =  34.1 %)
##
## Available components:
##
## [1] "cluster"      "centers"      "totss"        "withinss"     "tot.withinss"
## [6] "betweenss"    "size"         "iter"         "ifault"       "
```

Visualizing

```
fviz_cluster(kmeans_result, data = Pharmanumericals_normal)
```



The cluster plot gives two clusters with the sizes 11 and 10.

```
wss_clusters<-kmeans_result$cluster
dfl<-cbind(Pharmaceuticals_data,wss_clusters)

wss_table<-aggregate(dfl[, -c(1:2,12:14)], by=list(dfl$wss_clusters),FUN="median")
print(wss_table[, -1])
```

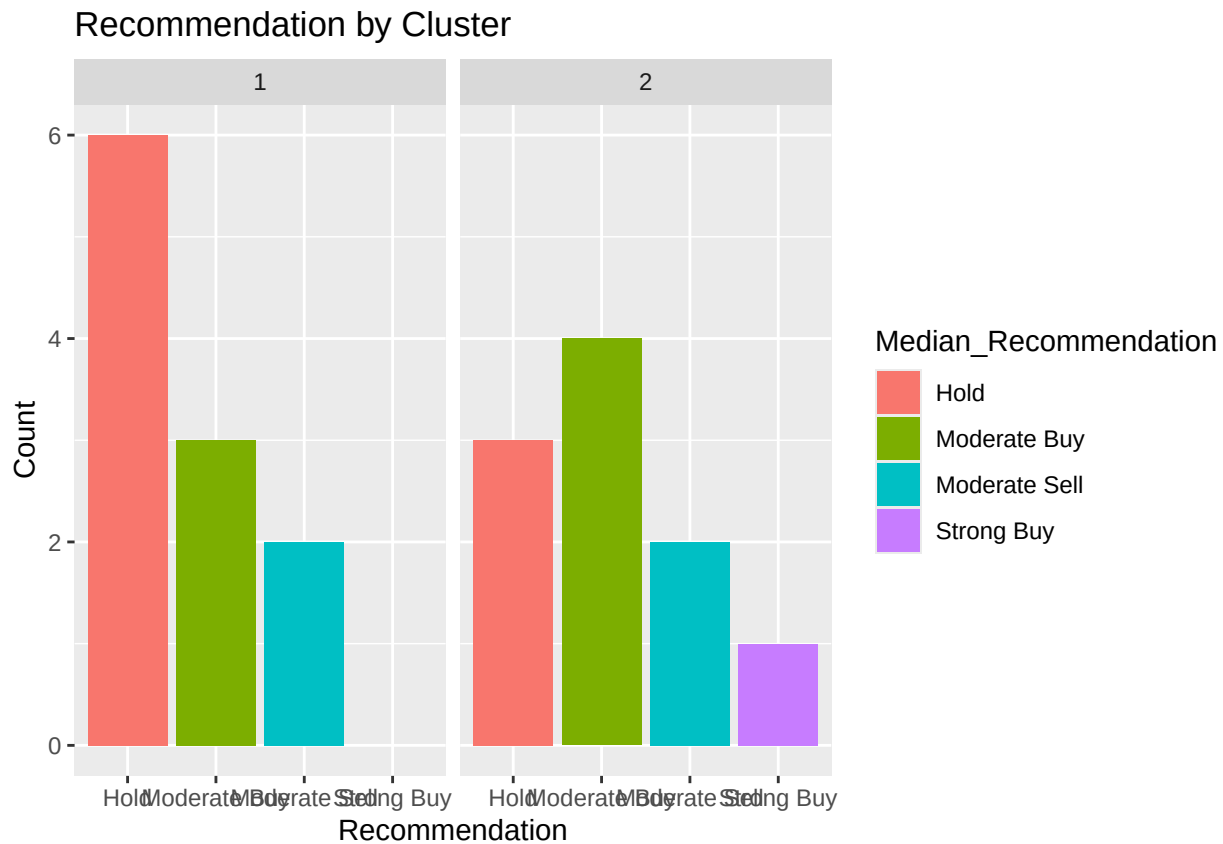
```
##   Market_Cap  Beta PE_Ratio  ROE  ROA Asset_Turnover Leverage Rev_Growth
## 1      73.84 0.460   21.50 31.0 15.0           0.8   0.280      8.560
## 2       4.78 0.555   23.35 14.2  5.6           0.6   0.475     14.495
##   Net_Profit_Margin wss_clusters
## 1              20.6             1
## 2              11.1             2
```

Recommendations

```
wss_summary <- table(dfl$wss_clusters, dfl$Median_Recommendation)
names(dimnames(wss_summary)) <- c("Cluster", "Recommendation")
wss_summary <- addmargins(wss_summary)
wss_summary
```

```
##           Recommendation
## Cluster Hold Moderate Buy Moderate Sell Strong Buy Sum
##      1      6          3          2          0  11
##      2      3          4          2          1  10
##      Sum      9          7          4          1  21
```

```
ggplot(dfl, aes(x = Median_Recommendation, fill = Median_Recommendation)) +
  geom_bar() +
  facet_wrap(~ wss_clusters) +
  labs(title = "Recommendation by Cluster", x = "Recommendation", y = "Count")
```

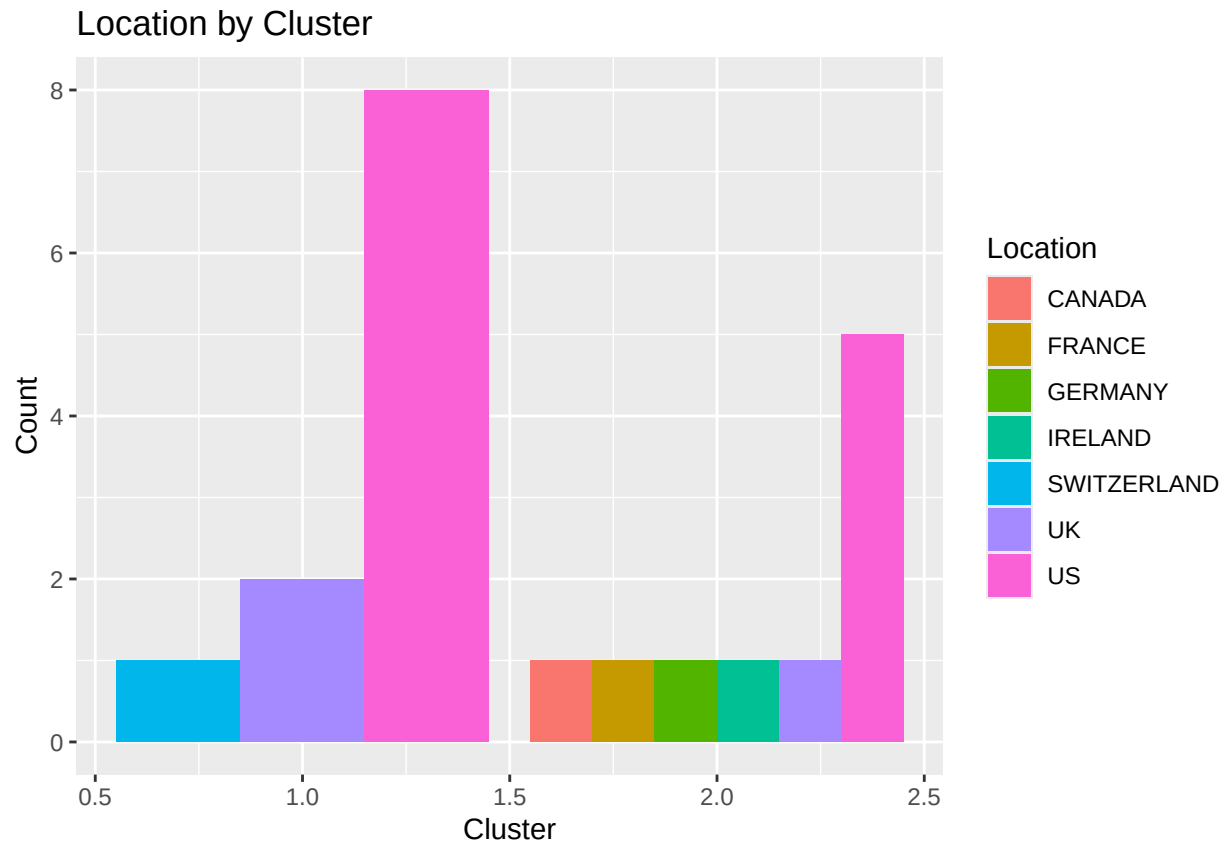


Locations

```
wss_location <- table(dfl$wss_clusters, dfl$Location)
names(dimnames(wss_location)) <- c("Cluster", "Location")
wss_location <- addmargins(wss_location)
wss_location
```

```
##      Location
## Cluster CANADA FRANCE GERMANY IRELAND SWITZERLAND UK US Sum
##      1      0      0      0      0      1  2  8  11
##      2      1      1      1      1      0  1  5  10
##      Sum      1      1      1      1      1  3 13  21
```

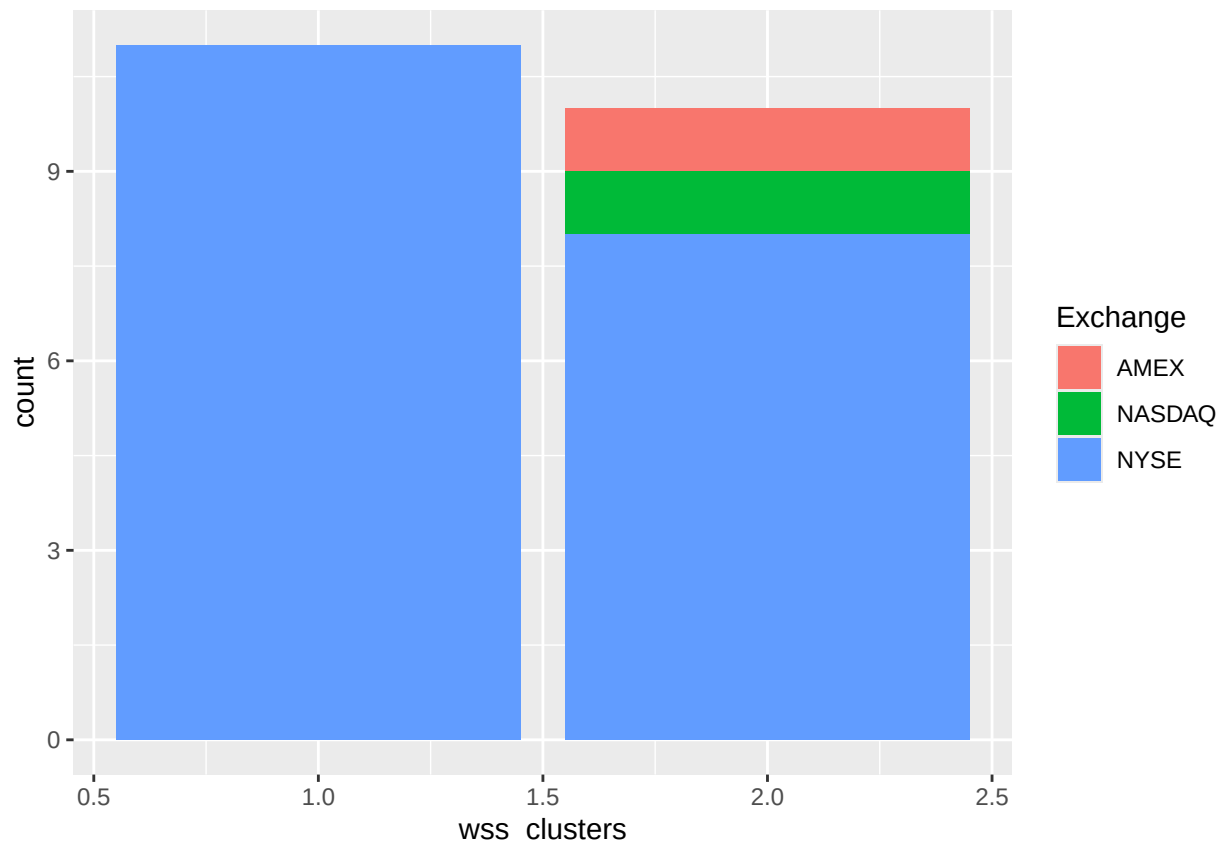
```
ggplot(dfl, aes(x = wss_clusters, fill = Location)) +
  geom_bar(position = "dodge") +
  labs(title = "Location by Cluster", x = "Cluster", y = "Count")
```



```
wss_exchange <- table(dfl$wss_clusters, dfl$Exchange)
names(dimnames(wss_exchange)) <- c("Cluster", "Exchange")
wss_exchange <- addmargins(wss_exchange)
wss_exchange
```

```
##      Exchange
## Cluster AMEX NASDAQ NYSE Sum
##      1      0      0    11  11
##      2      1      1     8   9
##      Sum      1      1    19  21
```

```
ggplot(dfl, aes(x=wss_clusters, fill=Exchange))+geom_bar()
```



Analysis: 1. Both the elbow and Silhouette methods determined $k=2$. 2. The further analysis is made using the elbow method. 3. Recommendations: Cluster 1 shows that most firms have a hold recommendations and Cluster 2 has firms giving more moderate buy. The buy recommendations are “0” for cluster 1. 4. Location: Cluster 1 is dominated by US based firms with 8 from US, 2 firms from UK and 1 from Switzerland. Cluster 2 is more diverse. This shows the smaller growth phase pharma companies are wide spread across multiple countries. 5. All 11 firms in Cluster 1 are listed exclusively on NYSE. Cluster 2, most firms (8 out of 10) are also on NYSE, but 1 firm is listed on NASDAQ and 1 on AMEX.

Naming of Clusters

Cluster:1 Established Pharma companies: These firms are large, financially strong, highly profitable and stable with high market cap, strong ROE and ROA, low leverage and the stock prices look stable. Most of the firms are based of US and are listed on NYSE.

Cluster:2 Emerging Pharma companies The cluster has firms with low market cap but high revenue, leverage and lower profitability metrics. They are geographically diverse and receive moderate buy recommendations, indicating their growth potential in the future.